

The Web Forces

Concept & Theme: “LogiPulse: Last-Mile Urban Logistics Engine”

The Real-World Scenario

Urban logistics platforms (like Uber, DoorDash, or DHL) live and die by real-time web dashboards. Operational dispatchers need to track hundreds of active deliveries, route drivers, handle delay alerts, and reassign orders on the fly.

Participants are hired as founding engineers to build LogiPulse—a high-performance, real-time dispatcher Web App for a fast-growing urban delivery fleet.

Base Problem Statement (Day 1)

Build a single-page web dashboard for dispatchers that manages live order states, driver assignments, and delivery tracking. The app must feature:

Active Order Pipeline (CRUD & Filtering)

- Display incoming delivery orders with status tags (Unassigned, In Transit, Delayed, Delivered).
- Interactive search, filtering by status/priority, and sorting by ETA.

Interactive Dispatch Console

- A dynamic modal or form allowing dispatchers to create new delivery orders or manually reassign an order to an available driver.

Live Fleet Analytics Bar

- Dynamic, auto-calculating counters showing: Total Active Orders, On-Time Rate %, Unassigned Orders, and Active Drivers.

The Mid-Challenge Twist (Day 2 Drop)

“The Gridlock Incident & Live Webhook Spike”

Halfway through Day 2, a simulated real-world operational disruption hits:

The Scenario

A major traffic disruption causes sudden driver delays, and the external API feed begins firing webhook alerts with delayed order flags and occasionally corrupted/incomplete payload data.

The Task

Teams must build a resilient client-side or server-side handler:

- Safely validate incoming order payloads and filter out malformed data without crashing the UI.
- Automatically highlight affected “High Delay” orders in red and pop up a real-time Dispatcher Action Banner prompting one-click driver reassignment.

2-Day Schedule & Time Distribution

The module runs as two focused build sessions — a 120-minute core build on Day 1 and a 90-minute crisis-response sprint on Day 2. Each session is broken into short, timed phases so every team moves through the same stages at the same pace:

Day 1 — Core Product Engineering (120 minutes)

Time	Duration	Phase	What Happens
00:00 – 00:15	15 min	PRD Briefing & Architecture Setup	Teams receive the full brief, ask clarifying questions, and plan their data model and architecture before writing any code.
00:15 – 01:15	60 min	UI Layout & Interactive Pipeline Feed	Teams build the dashboard layout and the live Active Order Pipeline — status tags, search, filtering, and ETA sorting.
01:15 – 02:00	45 min	Dispatch Logic & Dynamic Analytics Engine	Teams implement the Interactive Dispatch Console and wire up the Live Fleet Analytics Bar's auto-calculating counters.

Day 2 — Crisis Patching & Pitch (90 minutes)

Time	Duration	Phase	What Happens
00:00 – 00:10	10 min	Twist Drop: Gridlock Webhook Simulation	Organizers release the Day 2 twist; teams review the incoming corrupted webhook payloads and plan their fix.
00:10 – 00:45	35 min	Error Handling & Auto-Reassignment UI	Teams validate and filter malformed payloads, highlight “High Delay” orders in red, and build the one-click reassignment banner.
00:45 – 01:00	15 min	Production Deployment & README Spec	Teams deploy their live URL and write the Engineering Patch Log section of the README.
01:00 – 01:30	30 min	3-Minute Live Pitch & Judge Defense	Each team demos normal operations and the gridlock emergency, then takes technical questions from judges.

Total module time: 3 hours 30 minutes across both days (120 minutes on Day 1 + 90 minutes on Day 2).

Participant Deliverables & Technical Stack

Technical Requirements

- **Frontend:** Any modern web stack (React, Next.js, Vue, or native HTML5/Tailwind/JavaScript).
- **Data Layer:** Local state, Node/Express backend, Supabase/Firebase, or well-structured LocalStorage mock data layer.
- **Deployment:** Public live URL via Vercel, Netlify, GitHub Pages, or Render.

Final Deliverables

- **Production Web Application:** Live, fully functional URL.
- **GitHub Repository:** Public code repository with clean commits across both days.
- **Engineering Patch Log:** A concise section in README.md explaining how the team sanitized corrupted payloads and handled the traffic incident.
- **Live Product Pitch:** A 3-minute live demonstration of the dashboard handling normal operations and the gridlock emergency.

Judging & Evaluation Rubric (100 Points)

Criteria	Points	Evaluation Focus
Product Usability & UI/UX Design	35	Is the dashboard intuitive for a dispatcher? Is the layout clean, responsive, and easy to navigate under pressure?
Resilience & Twist Implementation	25	How seamlessly does the app catch malformed data payloads and highlight delayed orders without breaking?
Code Structure & Engineering Quality	20	Modular component design, clean state management, active Git history, and proper error handling.
Product Pitch & Live Demo	20	Clarity of the 3-minute demo, team coordination, and confidence answering judge technical Q&A.
Total	100	